Xiamen University Malaysia - EEE221 Signals and Systems

Tutorial 5: Fourier Transform (April 2026)

Name: [redacted] Student ID: [redacted]

Q1. Find the Fourier transform of x(t) = e^(-2t) u(t). Sketch its magnitude spectrum.

Q2. Using the time-shift property, find the transform of x(t - 3) for x(t) in Q1.

Q3. A signal has X(w) = 1/(1 + jw). Find the energy of x(t) using Parseval's theorem.

Q4. Show that the transform of a real, even signal is real and even.

Q5. Find the inverse transform of X(w) = 2 sin(w)/w and sketch x(t).